INCH-POUND

MS19070B
15 August 2011
SUPERSEDING
MS19070A
7 October 1987

DETAIL SPECIFICATION SHEET

# WASHER, KEY RETAINING, BALL AND ROLLER BEARING, REGULAR SERIES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-W-21338.

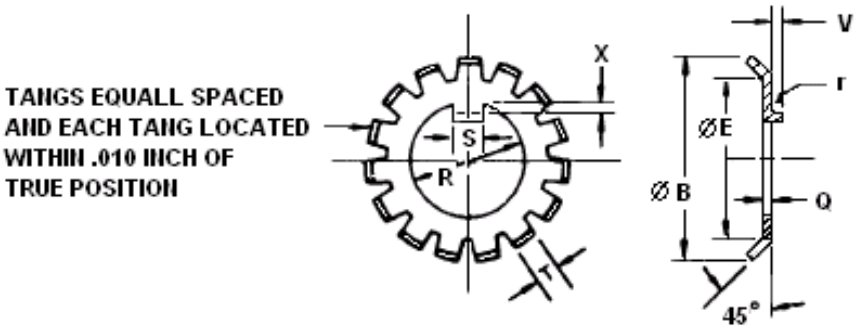



FIGURE 1. WASHER.

AMSC N/A

FSC 5310

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TABLE I. Dash Numbers and Dimensions.

| DASH NO | Q THICKNESS | | R BORE | | ∅E | | ∅B OVER TANGS MAX | KEY | | | | | | | TANGS | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | S WIDTH | | X DEPTH | | V PROJECTION | | r | NO. | T WIDTH | |
| | Q | TOL ± | MAX | MIN | E | TOL+ | | MAX | MIN | MAX | MIN | V | TOL+ | RAD | | T | TOL± |
| 00 | .042 | .004 | .421 | .406 | .625 | .015 | .875 | .120 | .110 | .030 | .020 | .062 | .031 | .016 | 9 | .115 | .005 |
| 01 | .042 | .004 | .499 | .484 | .719 | .015 | 1.016 | .120 | .110 | .030 | .020 | .062 | .031 | .016 | 9 | .115 | .005 |
| 02 | .042 | .004 | .616 | .601 | .812 | .015 | 1.156 | .120 | .110 | .030 | .020 | .062 | .031 | .016 | 11 | .115 | .005 |
| 03 | .042 | .004 | .694 | .679 | .938 | .015 | 1.328 | .120 | .110 | .030 | .020 | .062 | .031 | .016 | 11 | .115 | .005 |
| 04 | .042 | .004 | .816 | .801 | 1.125 | .015 | 1.531 | .176 | .156 | .030 | .020 | .062 | .031 | .016 | 11 | .156 | .010 |
| 05 | .050 | .004 | 1.009 | .989 | 1.281 | .015 | 1.719 | .176 | .156 | .030 | .020 | .094 | .031 | .016 | 13 | .156 | .010 |
| 06 | .050 | .004 | 1.213 | 1.193 | 1.500 | .015 | 1.922 | .176 | .156 | .050 | .035 | .094 | .031 | .031 | 13 | .156 | .010 |
| * 065 | .050 | .004 | 1.353 | 1.333 | 1.812 | .015 | 2.250 | .176 | .156 | .050 | .035 | .094 | .031 | .031 | 15 | .156 | .010 |
| 07 | .050 | .004 | 1.416 | 1.396 | 1.812 | .015 | 2.250 | .176 | .156 | .050 | .035 | .094 | .031 | .031 | 15 | .156 | .010 |
| 08 | .058 | .005 | 1.603 | 1.583 | 2.000 | .030 | 2.469 | .290 | .250 | .050 | .035 | .094 | .031 | .031 | 15 | .219 | .015 |
| 09 | .058 | .005 | 1.818 | 1.792 | 2.281 | .030 | 2.734 | .290 | .250 | .050 | .035 | .125 | .031 | .031 | 17 | .219 | .015 |
| 10 | .058 | .005 | 2.017 | 1.992 | 2.438 | .030 | 2.922 | .290 | .250 | .050 | .035 | .125 | .031 | .031 | 17 | .219 | .015 |
| 11 | .063 | .005 | 2.207 | 2.182 | 2.656 | .030 | 3.109 | .290 | .250 | .050 | .035 | .125 | .031 | .031 | 17 | .219 | .015 |
| 12 | .063 | .005 | 2.425 | 2.400 | 2.844 | .030 | 3.344 | .290 | .250 | .070 | .055 | .125 | .031 | .047 | 17 | .219 | .015 |
| 13 | .063 | .005 | 2.613 | 2.588 | 3.062 | .030 | 3.578 | .290 | .250 | .070 | .055 | .125 | .031 | .047 | 19 | .219 | .015 |
| 14 | .063 | .005 | 2.816 | 2.791 | 3.312 | .030 | 3.828 | .290 | .250 | .070 | .055 | .188 | .062 | .047 | 19 | .219 | .015 |
| 15 | .072 | .005 | 3.003 | 2.973 | 3.562 | .030 | 4.109 | .290 | .250 | .070 | .055 | .188 | .062 | .047 | 19 | .313 | .015 |
| 16 | .072 | .005 | 3.207 | 3.177 | 3.844 | .030 | 4.375 | .353 | .313 | .070 | .055 | .188 | .062 | .047 | 19 | .313 | .015 |
| 17 | .072 | .005 | 3.425 | 3.395 | 4.031 | .030 | 4.625 | .353 | .313 | .070 | .055 | .188 | .062 | .047 | 19 | .313 | .015 |
| 18 | .094 | .006 | 3.612 | 3.582 | 4.281 | .045 | 4.938 | .353 | .313 | .070 | .055 | .188 | .062 | .047 | 19 | .313 | .015 |
| 19 | .094 | .006 | 3.830 | 3.800 | 4.562 | .045 | 5.219 | .353 | .313 | .070 | .055 | .188 | .062 | .047 | 19 | .313 | .015 |
| 20 | .094 | .006 | 4.018 | 3.988 | 4.812 | .045 | 5.500 | .353 | .313 | .085 | .065 | .250 | .062 | .047 | 19 | .313 | .015 |
| 21 | .094 | .006 | 4.222 | 4.192 | 5.000 | .045 | 5.703 | .353 | .313 | .085 | .065 | .250 | .062 | .047 | 19 | .375 | .015 |
| 22 | .125 | .007 | 4.425 | 4.395 | 5.281 | .045 | 6.062 | .353 | .313 | .085 | .065 | .250 | .062 | .047 | 19 | .375 | .015 |
| 24 | .125 | .007 | 4.831 | 4.801 | 5.688 | .045 | 6.469 | .353 | .313 | .085 | .065 | .250 | .062 | .047 | 19 | .375 | .015 |
| 26 | .125 | .007 | 5.226 | 5.191 | 6.188 | .045 | 7.031 | .435 | .375 | .105 | .085 | .250 | .062 | .062 | 19 | .500 | .020 |
| 28 | .125 | .007 | 5.617 | 5.582 | 6.531 | .045 | 7.438 | .590 | .500 | .105 | .085 | .250 | .062 | .062 | 19 | .500 | .020 |
| 30 | .156 | .008 | 6.018 | 5.983 | 7.062 | .060 | 8.062 | .590 | .500 | .105 | .085 | .312 | .062 | .062 | 19 | .500 | .020 |
| 32 | .156 | .008 | 6.424 | 6.389 | 7.438 | .060 | 8.438 | .590 | .500 | .105 | .085 | .312 | .062 | .062 | 19 | .500 | .020 |
| 34 | .156 | .008 | 6.799 | 6.764 | 8.031 | .060 | 9.062 | .715 | .625 | .105 | .085 | .312 | .062 | .062 | 19 | .500 | .020 |
| 36 | .156 | .008 | 7.206 | 7.171 | 8.375 | .060 | 9.438 | .715 | .625 | .105 | .085 | .312 | .062 | .062 | 19 | .625 | .020 |
| 38 | .156 | .008 | 7.612 | 7.577 | 8.781 | .060 | 9.875 | .715 | .625 | .105 | .085 | .312 | .062 | .062 | 19 | .625 | .020 |
| 40 | .156 | .008 | 8.017 | 7.982 | 9.156 | .060 | 10.312 | .840 | .750 | .105 | .085 | .312 | .062 | .062 | 19 | .625 | .020 |

* Inactive for new design after 1987.

NOTES:

1. MATERIALS:
   (a) Carbon steel, AISI 1010 to 1035 (UNS G10100 to G10350), or 1116 to 1118 (UNS G11160 to G11180).

   (b) Corrosion resisting steel AISI 302, 303 or 304 (UNS S30200, S30300, or S30400).

2. TEMPER: Each washer shall be of such temper as to permit its being bent 180”, without fracture, either way of the grain, over a radius equal to one half the stock thickness.

3. PROTECTIVE COATING:
   (a) Carbon steel washers are available either zinc plated or unplated. Zinc plated washers shall be plated in accordance with ASTM B633, Class 5, Type II. Cadmium plating in accordance with SAE-AMS-QQ-P-416, Type II, Class 3 shall no longer be furnished.

   (b) Corrosion resisting steel washers shall be passivated in accordance with SAE-AMS2700, Method I, Type 2.



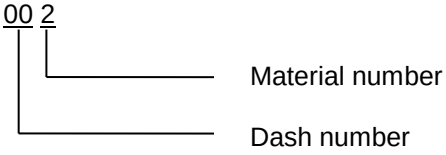
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4. DIMENSIONS: All dimensions are in inches. Tolerances: decimals ± .010, angles ±2° unless otherwise specified.

5. PART NUMBER: The MS part number is the MS number, plus the dash and the material number. The material numbers are:

   1 - unplated carbon steel
   2 - zinc plated carbon steel
   3 - corrosion resisting steel

   Example: MS19070 – 00 2




6. NUTS: See MS19068 for applicable nuts. Applicable nuts should have size numbers corresponding to washer dash numbers: e.g. washer size 00 should be used with nut size number 00.

7. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

8. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - OS
Air Force - 11

Review activities:
Navy – AS, SA

Preparing activity:
DLA - IS

(Project 5310-2011-007)

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